(1) (b) Small k with noisy data
(2) (c) They always overfit
(3) (d) Improving feature selection
(4) (c) All features are considered at each split
(5) (c) Features are Independent
(6) (c) Sigmoid
(7) (c) Accuracy
(8) (d) Overfitting
(9) (c) Because distance calculation depends on scale
(10) (c) Logistic Regression
~~(11)~~
~~(12)~~
~~(13)~~
~~(14)~~
~~(15)~~

Q13 Sol →

	Predicted Fraud	Predicted Not
Actual Fraud	120 TP	30 FN
Actual Not Fraud	50 FP	800 TN

Sol → (a) Accuracy = ~~$\frac{TP + TN}{Total}$~~ $\frac{TP + TN}{TP + TN + FP + FN}$

$\Rightarrow \frac{120 + 800}{120 + 30 + 50 + 800} = \frac{920}{1000}$

→ ~~0.92~~ 92%

(b) Precision :→ $\frac{TP}{TP + FP}$

$\Rightarrow \frac{120}{120 + 50} = \frac{120}{170} = \ldots = \ldots = 70\%$

0.8
5)40
40

(c) Recall :- $\frac{TP}{TP+FN}$

$\Rightarrow \frac{120}{120+50} = \frac{12}{17} \Rightarrow \frac{120}{170} = 80\%$

(d) F1 Score :- $2 \times \frac{P \times R}{P+R}$

$\Rightarrow 2 \times \frac{92 \times 70}{92+70}$

$\Rightarrow \frac{2 \times 1.2}{2.3}$

$\Rightarrow \frac{2.4}{2.3} \Rightarrow \frac{24 \times 10}{23 \times 10}$

$\Rightarrow 75\%$

Q → From the above Calculation Recall has 80% Because 20% model misses actual frauds. This model is not acceptable.

Q11 Overfitting in Decision Tree :-

Overfitting happens when the tree becomes too deep & Complex. Overfitting in decision Tree is Connected to depth tree because depth tells how the model is Complex

Depth means how many splits the tree is allowed to make from root to leaf

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How to prevent overfitting?

We can prevent overfitting by using

(i) max_depth → how deep the tree will grow

(ii) Pruning → Removing unnecessary Branches

Bagging :- It creates many decision Trees & average their prediction. In this we take samples from the dataset. Train a seperate deep tree on each sample And Combine prediction using voting or averaging. In this each tree overfit in different ways. For eg :- random noise cancels out, prediction become stable; Common patterns remains.

Random Forest :- Random Forest = Bagging + extra Randomness.

In addition to random data, it also uses random features. When splitting a node, Bagging tree can use all features; Random Forest can use only a random subset of a features.

→ Bagging reduces overfitting by averaging many overfit trees while Random Forest adds extra randomness so the trees are less similar, making the average even reliable.

Q13

(1)

(2)

IMPORTANT NOTES

Q13 Random Forest :- In this
. There is Bagging + extra Randomness.
In addition to random data, it also use random features. when splitting a, Bagging tree can use all features, Random forest can use only a random subset of feature.

(1) Bootstrap sampling :-> It is sampling with replacement This means every time you pick the data point in Bootstrap then it Samples with the ~~replacement~~ you choose the right one into the original pool before picking the next one

(2) Random feature selection :-> It is a technique, where only random ~~feature~~ subset of available features is considered when building a model or making a model.